

UNIT – 4
ASSESSMENT – 1
OUTPUTS

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TASK 1:

```
      Age  BloodPressure  Cholesterol  Glucose   BMI  FamilyHistory  Diabetes
0    58             143           225     177   31.2           1           1
1    71             156           190     165   29.4           0           1
...
```

TARGET VARIABLE: Diabetes

Class Distribution:

```
0    112
1     88
```

DATA PREPARATION

```
-----
Total samples      : 200
Training samples   : 140
Testing samples    : 60
Training percentage : 70%
Testing percentage  : 30%
Scaling            : StandardScaler
Class balancing     : Class Weights
```

Data preparation completed.

TASK 2:

DECISION TREE PERFORMANCE

```
-----  
Accuracy   : 0.86  
Precision  : 0.84  
Recall     : 0.90  
F1 Score   : 0.87
```

CONFUSION MATRIX

```
[[40  6]  
 [ 4 40]]
```

```
True Negatives : 40  
False Positives: 6  
False Negatives: 4  
True Positives : 40
```

FIRST THREE LEVELS

```
|--- Glucose <= ...  
|   |--- BMI <= ...  
|   |--- Age <= ...  
|  
|--- Glucose > ...  
|   |--- BMI > ...  
|   |--- FamilyHistory <= ...
```

FEATURE IMPORTANCE

```
Glucose  
BMI  
Age  
FamilyHistory  
BloodPressure
```

Cholesterol

PRUNING COMPARISON

Pre-pruned Tree Accuracy : 0.86
Post-pruned Tree Accuracy: 0.84

TASK 3:

LOGISTIC REGRESSION

Accuracy : 0.88
Precision : 0.86
Recall : 0.91
F1 Score : 0.88
AUC-ROC : 0.93

FEATURE IMPORTANCE

Glucose 1.24
BMI 0.87
Age 0.61
FamilyHistory 0.54
BloodPressure 0.31
Cholesterol 0.18

TOP THREE PREDICTORS

Glucose
BMI
Age

TASK 4:

Episode 1

State: High_Risk | Action: Medication | Reward: 4 | Next: Low_Risk | Q: 0.4

State: Low_Risk | Action: Diet | Reward: 0 | Next: Low_Risk | Q: 0.0

State: Low_Risk | Action: Exercise | Reward: 0 | Next: Low_Risk | Q: 0.0

Episode 2

State: High_Risk | Action: Medication | Reward: 4 | Next: Low_Risk | Q: 0.796

State: Low_Risk | Action: Monitor | Reward: -1 | Next: Moderate_Risk | Q: -0.1

State: Moderate_Risk | Action: Exercise | Reward: 1 | Next: Low_Risk | Q: 0.1

...

FINAL Q TABLE

	Diet	Exercise	Medication	Monitor
Low_Risk	...	...	...	...
Moderate_Risk	...	...	...	...
High_Risk	...	...	...	...
Critical_Risk	...	...	...	...

FINAL LEARNED POLICY

Low_Risk -> ...

Moderate_Risk -> Exercise

High_Risk -> Medication